

LESSON 2.4: Mailbox Management

What this lesson is about

This lesson provides a high-level overview of the mailbox system in Sterling B2B Integrator. to manage, create the password, user account, permission, and groups. Also explains about roles-based security and system news.

What you should be able to do

After completing this lesson, you should be able to:

- Describe Sterling B2B Integrator mailboxes.
- Add and view data in a mailbox.
- Extract data from a mailbox.
- Describe the impact of mailbox in Sterling B2B Integrator performance.

How you will check your progress

- The progress of the lesson is analyzed based on the successful application of the topics in the scenario and test the exercise independently.

References

Documentation

- Mailboxes:
http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/com.ibm.help.webdav_server.doc/SI_setup_mailbox.html
 - Global High Availability mailbox:
http://www.ibm.com/support/knowledgecenter/SS3JSW_5.2.0/nav_container_topics/sub_overview_system_admin_glbl_mbx.html
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Sterling B2B Integrator Mailboxes

Introduction

The Sterling B2B Integrator Mailbox system is a secure business document repository with a store-and-forward communication infrastructure. The Sterling B2B Integrator Mailbox adds to the business automation and facilities of Sterling B2B Integrator to conveniently build B2B electronic commerce communities. In addition, the Sterling B2B Integrator Mailbox provides a trading partners browser interface that is called the Mailbox Browser Interface (MBI). No additional software is required when the Sterling B2B Integrator Mailbox is deployed to an electronic trading community. The MBI is secure and simple to use, requiring no special user training.

Components

The Sterling B2B Integrator Mailbox system that is managed by Sterling B2B Integrator includes the following components:

- MBI
- Messages
- Mailbox
- Archive

Mailbox Browse Interface(MBI)

The MBI allows Internet users to securely access their mailboxes from a standard web browser. Therefore, the MBI is a web application and requires only a browser. No proprietary client software is required.

Users can send and retrieve messages to and from, and search messages in any mailbox for which they have permissions. The MBI allows users to change their account password.

Messages

Business documents that are stored in the Sterling B2B Integrator Mailbox are called *messages* and can contain business content in any format such as binary, EDI, or XML. Each message is assigned an extraction policy that specifies the rules for extracting messages from a mailbox. A message added to a mailbox can be processed immediately or at a scheduled time in the future.

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Sterling B2B Integrator Mailboxes

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Mailbox

A mailbox is a storage area for business documents and provides an administrative hierarchy that is easy to manage and understand.

Mailboxes form a hierarchy. The top of the hierarchy is called the root mailbox. A slash (/) denotes the root mailbox. Mailboxes are able to support submailboxes. This organizational concept is known as mailbox hierarchy.

System users have access to their documents while administrators can organize and manage documents across all mailboxes.

Managing Mailbox Messages

Introduction

A key function of Sterling B2B Integrator is the routing of files from one mailbox to another. Sterling B2B Integrator uses the mailbox system within the Sterling B2B Integrator application for its routing of messages.

The Sterling B2B Integrator Mailbox system manages messages by using routing rules and services. Routing rules are used to manage business document processing. Routing rules are used to schedule processing the business document and can be configured for automatic, scheduled, or manual processing of the message. Services are used to add messages to a particular mailbox, locate messages, delete messages, extract messages, and schedule the routing rule evaluation.

Mailbox creation

Mailboxes that are used for Sterling B2B Integrator operations can be created automatically in the following circumstances:

- When a Partner is created in Sterling B2B Integrator
- When Partners are migrated from Advanced File Transfer
- When Partners are imported into Sterling B2B Integrator
- When a Routing Channel is created
- At file transfer run time (by dynamic routing channel templates)

When a Partner is created, a mailbox with the same name is created. For example, if the partner is named "PartnerOne" a mailbox is created where the mailbox absolute path is "/PartnerOne". When the PartnerOne user logs in, the absolute path is not displayed. The PartnerOne user sees the virtual path only, which is in this example "/".

When a routing channel is created, a producer mailbox is created in the structure that is specified in the governing Routing Channel Template. Creation of a Routing Channel need not create a consumer mailbox. A consumer mailbox can be created on demand (at run time) if configured to do so in the Routing Channel Template. Whether created at Routing Channel creation time or on demand at run time, the consumer mailbox conforms to the structure mandated by the Routing Channel Template. Managing Mailbox Messages

Sterling B2B Integrator can also use pre-existing mailboxes (for example, from a Sterling B2B Integrator instance) if their paths are the ones a Routing Channel expects. The routing channel derives its mailbox path from the governing Routing Channel Template.

For Routing Channels, Sterling B2B Integrator checks to determine whether a mailbox of the required structure exists. The Sterling B2B Integrator creates the mailbox along with any required permissions if it does not exist.

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Managing Mailbox Messages

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Routing Rules

Routing rules provide a powerful and easy-to-use mechanism for controlling document routing. Additionally, the Sterling B2B Integrator Mailbox is easily combined with Sterling B2B Integrator capabilities such as business processes, document translation, B2B protocols, and backend system integration.

You can create routing rules to initiate action automatically when a message is added to a mailbox. Automatic routing assures that the rule is evaluated. When a message is added to a mailbox, any rules that are established for the Sterling B2B Integrator Mailbox are evaluated automatically. Later, any matching rules are executed. In some situations, as a result of an outage, a message can be routed before the outage and again at system restart.

Manual and scheduled evaluations do not provide a message that is delivered once. Routing usually occurs, but if a system outage occurs, the routing does not resume at system start.

Routing Rule Pattern and Action

Routing rules consist of two parts: a pattern to specify when the rule applies, and an action the rule must perform. Routing rules contain string patterns that are used to match message names. This string can contain wildcard characters. You can restrict the application of the rule to specific mailboxes. When a message name matches the string pattern of a rule, the rule triggers the execution of a designated Sterling B2B Integrator business process. That process can perform any processing, including extraction of the message and integration with backend systems or a notification action. For example, a routing rule can trigger a business process that pages an administrator whenever a message arrives in the Dead Letter mailbox.

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Managing Mailbox Messages

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Routing Rule Evaluation mode

A routing rule evaluation mode is used to define when the rule is evaluated. You establish the evaluation mode when you define the routing rule. The following list describes the rule evaluation modes:

- Manual: The rule is evaluated when an administrator request evaluation (Test mode). The same batch of message ID is pulled each time that the rule is started.
- Scheduled- A schedule is established to evaluate rules. The message ID is pulled upon success request.

When a routing rule is evaluated, the Sterling B2B Integrator Mailbox searches the mailboxes that are defined in the rule for message names that match the pattern that is defined in the rule. When a match is found, the routing rule notifies the business process or contract that is specified in the routing rule. It depends how the business process is defined, it can extract and process the message, extract the message, and pass it to another application for processing, or notify another application to extract and process the message. If routing rules are set to be evaluated manually, a user or a business process can evaluate it.

Global mailbox

Introduction

Global High Availability Mailbox is a software component that enables your application servers to work together over multiple data centers in different locations. If a data center is unavailable, the remaining data centers continue processing transactions and managing the data.

The main tasks of Global Mailbox are:

Administering Global Mailbox to manage users, user permissions, virtual roots, events, mailboxes, and message.

Securing Global Mailbox to secure resources, users, communication, and access.

Monitoring Global Mailbox to monitor mailbox contents and properties either from within Global Mailbox or from another application that is registered in Global Mailbox.

Performance and tuning for Global Mailbox to tune performance to meet the needs during and after the installation of Global Mailbox.

Troubleshooting and support for Global Mailbox to isolate and resolve problems with Global Mailbox.

Exercise 2.4.1: Local mailbox

Introduction

Angela needs to locate the mailbox available in Sterling B2B Integrator and verify the routing rules configurations.

Instructions

Access the B2B Console (Sterling B2B Integrator) to locate the mailboxes that are created as a result of the Routing Channel creation.

Step 17: From the **Tools** menu, select **B2B Console**.

Step 18: From **Administration Menu**, select **Deployment > Mailboxes > Configuration**.

Step 19: From the **Mailbox Configuration** menu, use the **Go!** button under the **List** option to list all mailbox configurations.

Step 20: View the list of mailboxes.



Note

Make sure that Pop-up Blocker is off.

Enable the routing rule

To enable the routing of files from one mailbox to another, you must ensure that an appropriate mailbox routing schedule is enabled. The two routing rules that are installed with Sterling B2B Integrator:

- **MailboxEvaluateAllAutomaticRules**- runs every minute and can be edited for longer intervals.
- **MailboxEvaluateAllAutomaticRulesSubMin**- checks for the presence of routable messages every 10 seconds and can be edited for other intervals of less than 1 minute by modifying the MailboxEvaluateAllAutomaticRulesSubMin business process.

Use the following steps to ensure that a routing schedule is enabled

Step 1: From the **Tools** menu, select **B2B Console**.

Step 2: From **Administration Menu**, select **Deployment > Schedules**.

Step 3: Click **Go!**

Step 4: Locate either **MailboxEvaluateAllAutomaticRules** or **MailboxEvaluateAllAutomaticRulesSubMin** and ensure that it is enabled.

Impact of mailboxes in performance

Introduction

To avoid performance issues, you need to perform the following important activities.

- Disabling Duplicate Messages
- Using Mailbox Scheduled Delete service



Note

The mailbox storage is not suitable for storing the large files as it can affect the performance of Sterling B2B Integrator.

Disable duplicate messages

A global system setting allows or disallows duplicate messages. By default Sterling B2B Integrator allows duplicate messages. You can disallow by tuning mailbox properties through **customer_overrides.properties**. For optimum system performance, disallow duplicates in the `customer_overrides.properties` file. This setting ensures that every message in a single mailbox has a unique name.

If a message is written to a mailbox and the message name matches the name of an existing message in the mailbox, the service deletes the old message and the new message.

To make this setting, you need to edit the `customer_overrides.properties` file.

Step 1: In the **install_dir/properties** directory, locate (or create if necessary) the **customer_overrides.properties** file.

Step 2: Open the **customer_overrides.properties** file in a text editor.

Step 3: Add the following parameter:

```
mailbox.disallowDuplicateMessages=true
```

Step 4: In the **install_dir/bin** directory run `hardstop.sh` and then run `setupfiles.sh`.

Step 5: Execute `run.sh` to start the Sterling B2B Integrator.

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Impact of mailboxes in performance

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Purge messages

The messages in consumer mailboxes are not automatically purged and over time they can negatively affect system performance. To avoid this issue, configure the Mailbox Scheduled Delete service to delete messages from one, many, or all mailboxes.

Configuring and Scheduling Mailbox Scheduled Delete Service

Consider a scenario that you need to schedule the Mailbox Scheduled Delete service daily at 11:55 PM.

To schedule the Mailbox Scheduled Delete service:

1. Select Deployment > Services > Configuration from the Administration Menu. The Services Configuration page is displayed.
2. Click Go! next to New Service under Create.
3. Enter Mailbox Scheduled Delete Service in the Service Type text box and click Next.
4. Enter Mailbox Scheduled Delete Service in the Name text box and enter the meaningful description.
5. Set the value as 20 for Delete only messages older than this number of days.
6. Accept the defaults and click Next.
7. Select the Run daily under Schedule.
8. Enter schedule time as 11:55 PM.
9. Click add to set the schedule and then click Next.
10. Click Next and review the settings.
11. Click Finish to create the schedule.
12. Go to Deployment > Schedules from the Administration Menu.
13. Enter the MailboxScheduledDeleteService next to Search by Name under Search and click Go!.

Result

You can view the scheduled Mailbox Scheduled Delete Service.



A business process is unexpired (either purge or archive) if it is associated with a workflow ID for a document that is stored in a mailbox. The business process maintains a default expiration date of the completion date plus 10 years until the document is deleted from the mailbox. The default process for removing documents from mailboxes. The Mailbox Delete service must be scheduled or regularly executed by the customer to remove older documents from mailboxes when they are no longer needed.

Lesson review

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What you have been able to do

After completing this lesson, you should have been able to:

- Describe Sterling B2B Integrator mailboxes.
 - Add and view data in a mailbox.
 - Extract data from a mailbox.
 - Describe the impact of mailbox in Sterling B2B Integrator performance.
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